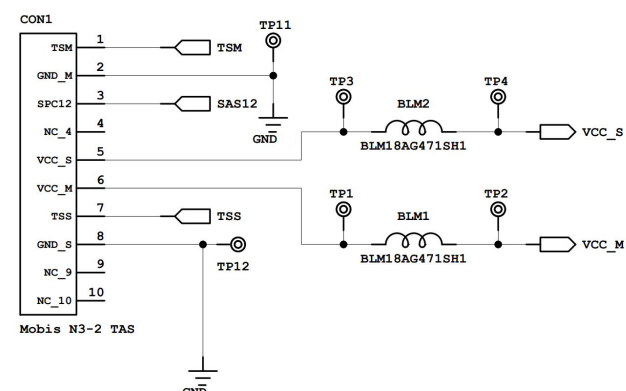
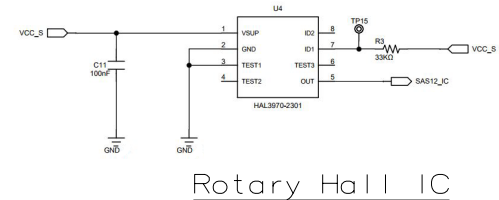
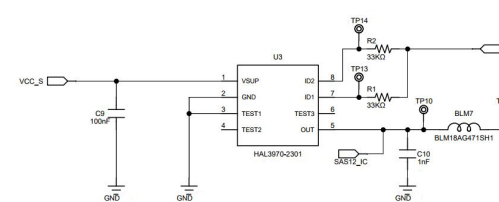
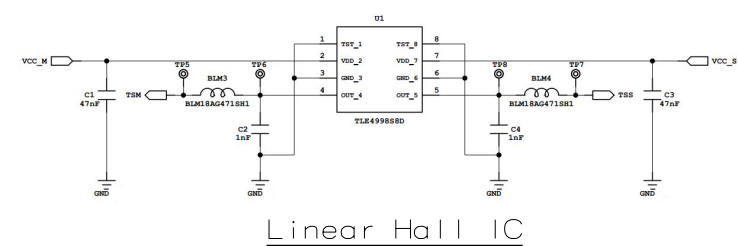
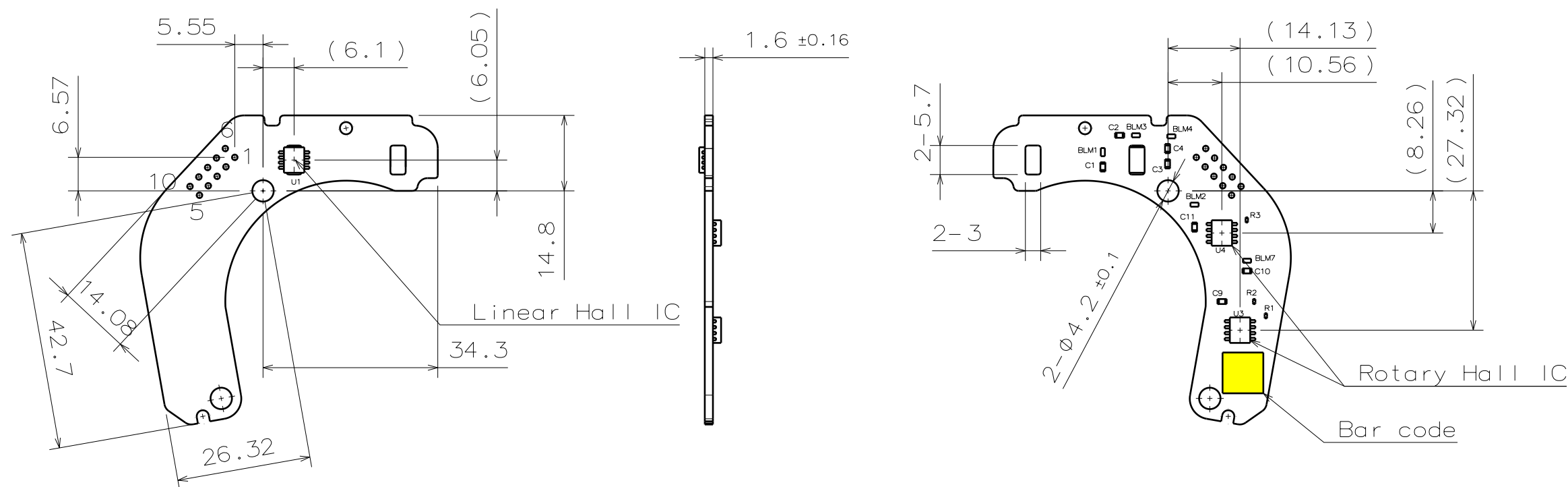


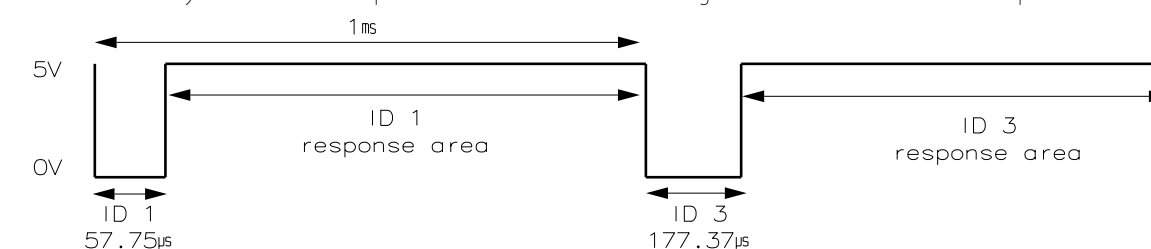


Pin No.	Description
1	TSM_1_SENT
2	GND
3	Angle_1_SPC
4	—
5	Vcc_S
6	Vcc_M
7	TSS_1_SENT
8	GND
9	—
10	—

PIN Map



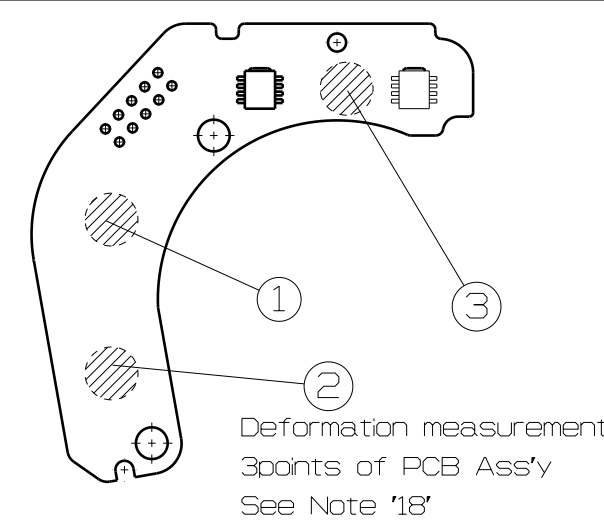
1. This is a PCB Assy's drawing.
2. The detail for circuit and pattern, follow the artwork.
 - PLM Document : 2025-PLM-D00310
3. All dimensions and tolerances of PCB follow the single pcb drawing(3PCR03624A).
4. No swap error, omission and misalignment of the chip component.
5. No PCB degradation, burn out, flux and alien substance after SMT.
6. No deformation and damage at PCB.
7. Coating liquid must not spill out on the side of the PCB.
8. Appearance criteria after coating follow the PCR Rev. 3
9. No assembly problem with Upper Case Assy
10. All components must be automotive qualified(AEC qualified)
11. Solder Paste : TAMURA TLF-204-HSP
12. Conformal coating materials : DOW CORNING 3-1953 or Chemtros PAS7900(A), (Type A-1(MS941-04))
13. Conformal coating area
 - Follow the left conformal coating area.
 - The surface mount of the circuit components
14. No conformal coating area
 - Follow the left conformal coating area.
 - Exception area of coating : 10pin terminal holes, Screw holes, Hall IC hole area, Assembly area with case
15. Conformal coating thickness : 30 ~ 150 μ m
16. Check coating area whether coating is applied on essential coating area or not.
17. All operators must wear a ESD protection bracelet
 - and must be equipped with ionizer in the main process(material warehouse, IC).
18. PCB Assy's bending quantity : Max 500 μ m/m(No crack all part)
19. ICT inspection items and specs should be selected after consultation with the LGIT quality manager.
20. All PCB Assy's must be inspected as the following 'Test item' and example of trigger pulse for SAS12 is below



21. Barcode position and type must be discussed with R&D.
22. Cleanliness Specification
- 22-1. Part Cleanliness – Metal : 400um↓, Non Metal : 600um↓
- 22-2. Box Cleanliness – Metal : 400um↓, Non Metal : 600um↓
23. This part must meet the following regulation.
- 1) ES90000-01 Rev. 10
 - 2) ES90000-02 Rev. 7
 - 3) ES90000-03 Rev. 6
 - 4) ES90000-04 Rev. 9
 - 5) MS201-02 Rev. 47
 - 6) MS915-01 Rev. 4 Type A
 - 7) MS941-01 Rev. 8 Type B
 - 8) MS941-04 Rev. 11 Type A-1
 - 9) MS184-01 Rev. 6 Type B
 - 10) POR Rev.3

24. Packaging container which reflects electro-static discharge measures should be used.
25. Each products should be free from interference.
26. Contaminants inflow and shock during movement should not be caused.
27. Following the below QR code rule
- Model [6:7] : A1

BYTE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
ITEM	Bare PCB LOT (YYWW)			Baking	Model	Supplier code					Line No.	YY	M	D	Serial No.			Rev.		



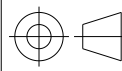
Test Item	Test Method	Acceptance Criteria
TOS output	Measure Unit time of SENT(TSM, TSS) after applying 5V at VCOM, VOCS.	3 μ s $\pm$ 20%
SAS output	Measure Unit time of SPC(SAS1, SAS2) after applying 5V at VOCS with Trigger pulse ID1 and ID3 at SAS12. Trigger pulse for ID1 : 57.75 μ s (52.25 μ s ~ 63.25 μ s) Trigger pulse for ID3 : 177.37 μ s (169.12 μ s ~ 185.62 μ s)	2.75 μ s $\pm$ 5%
TAS consumption _current	Measure consumption after applying 5V at VCOM, VOCS	Power A : 3~8mA Power B : 16~32mA

* Power A : TSM(T1) Power Line Pin No.6(VCC_M), No.2(GND)
Power B : TSS(T2), SAS1, SAS2 Power Line Pin No.5(VCC_S), No.8(GND)
2.2kΩ pull-up resistor must be applied on each output line.

DIMENSIONAL TOLERANCE	
~ up to 6	±0.1
over 6 up to 30	±0.2
over 30	±0.3

	C
	H
	A

	25.07.25	S.W.BYUN	Released
REV. NO.	DATE (YY MM DD)	SIGNATURE	CHANGE CONTENTS

0.0004kg	10	5REQ00109A		Coating agent		-	
0.0016kg	9	5SOC00024A		Solder cream		-	
3	8	2RSC00822A		Yageo, Chip resistor AC0402FR-0733XL, 33kohm, 100S		R1, R2, R3	
5	7	2FTC00096A		Murata, Ferrite bead BLM18AG471SH1		BLM1, 2, 3, 4 7	
2	6	2CAC01466A		SAMSUNG, Open type MLC CL10B104KB8WFC, 100nF, 50V, 160S		C9, C11	
3	5	2CAC01814A		SAMSUNG, Open type MLC CL10B102KB8WFC, 1nF, 50V, 160S		C2, C4, C10	
2	4	2CAC01803A		SAMSUNG, Open type MLC CL10B473KB8WFC, 47nF, 50V, 160S		C1, C3	
2	3	2SSF00026A		Micronas, Rotary Hall IC HAL3970-2301		U3, U4	
1	2	2ICZ00257A		Infineon, Linear Hall IC TLE4998S8D		U1	
1	1	3PCRO3624A		PCB FR-4, 2Layer, Middle-Tg, ENIG		-	
Q'TY	NO	PART NO		DESCRIPTION	MATERIAL	COLOR FINISH	NOTE
RELATED P/N				DESIGN 2024.11.22	TITLE		
		THIRD ANGLE PROJECT		S.W.BYUN	PCB Assy		
		1:1 mm		CHECKED 2024.11.22	PART NO		
				C.H.LEE	8PCA00019A		
				APPROVED 2024.11.22	MODEL		
				J.Y.KIM	ASCMFN0004		
					DWG NO		
					8PCA00019A		
 LG Innotek							